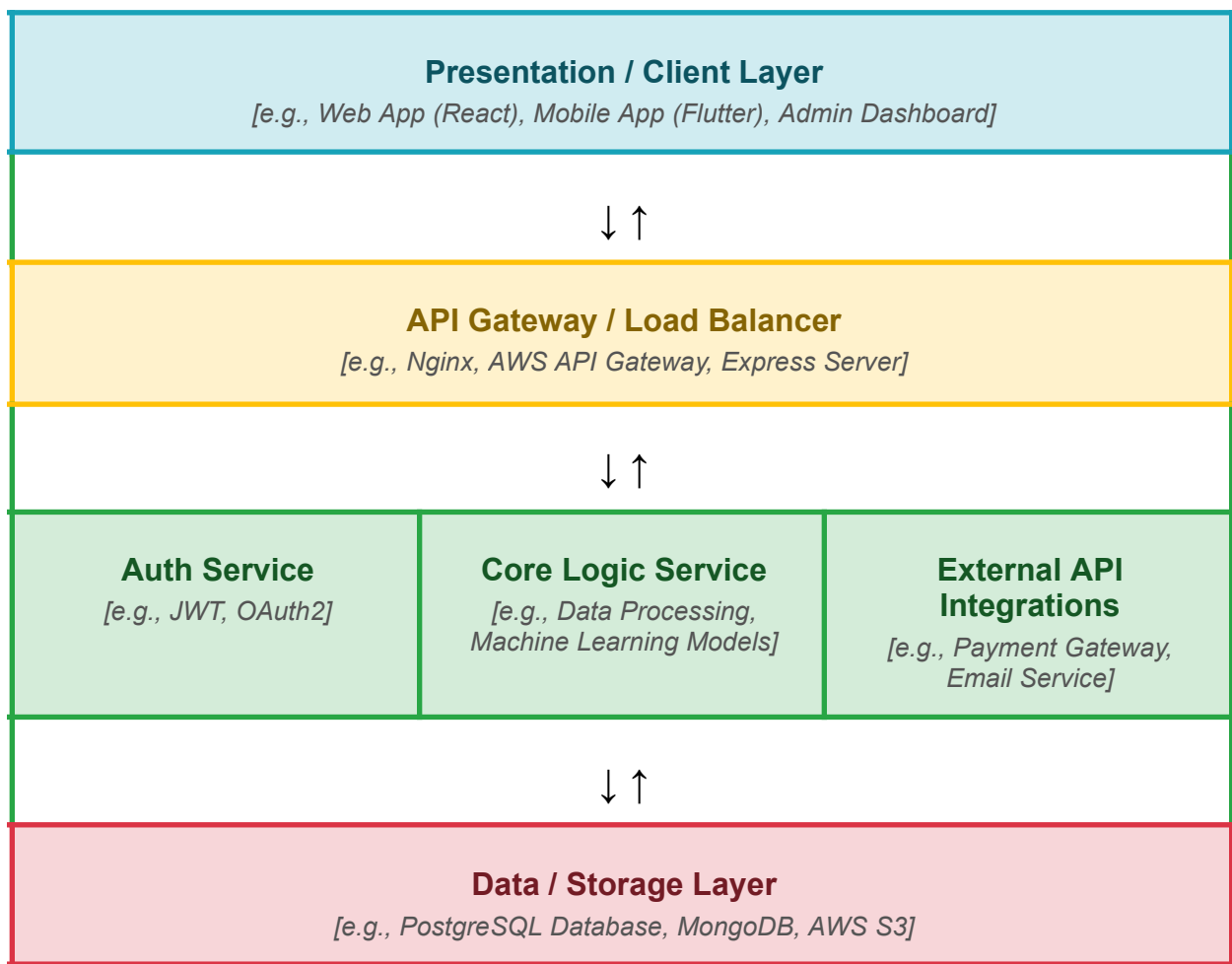


Solution Architecture

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	[Provides the web interface for students and teachers to upload audio, view results, and download reports.]	[Streamlit]
[API Gateway]	[Handles requests between the frontend and backend services.]	[Fast API]
[Microservices]	[Performs speech-to-text conversion, semantic similarity analysis, AI feedback generation, and report creation.]	[Whisper, Sentence Transformers, Google Gemini, ReportLab]

<i>[Database]</i>	<i>[Stores user information, transcripts, scores, and evaluation history.]</i>	<i>[SQLite, SQLAlchemy]</i>
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